

JAYANI SAMARAKOON

Undergraduate in Software Engineering

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PROFESSIONAL SUMMARY

Software Engineering undergraduate with hands-on experience in Machine Learning, AI, full-stack web development, mobile application development, and MLOps. Experienced in building end-to-end AI applications involving data preprocessing, feature engineering, predictive modeling, REST APIs, model deployment, and responsive user interfaces. Familiar with Python, Scikit-learn, PyTorch, TensorFlow, React.js, FastAPI, MLflow, DVC, Docker, and Apache Airflow. Seeking a Software Engineering / AI/ML internship to apply technical skills in real-world projects and contribute to a collaborative development team.

EDUCATIONAL QUALIFICATIONS

BSc (Hons) in Software engineering *2023-Present*

Sri Lanka Technology Campus (SLTC)

G.C.E. Advanced Level *Year of 2022*

K/MW Pinnwala central college

Technology stream: Engineering technology , Science for technology , Information Technology

SKILLS

Technical Skills

Programming	Python, Java, JavaScript, SQL
Frontend	React.js, HTML, CSS
Backend	FastAPI, REST APIs
Mobile	Flutter, Android Studio
Machine Learning	Scikit-learn, XGBoost, Random Forest, Logistic Regression
Deep Learning & AI	PyTorch, TensorFlow, Transformers, BERT, CLIP, OpenCV
MLOps	MLflow, DVC, Apache Airflow, Docker, DAGsHub
Databases	MySQL, Firebase
UI/UX	Figma, Wireframing, Prototyping
Tools	Git, GitHub, VS Code, Jira

Soft Skills

- **Team Collaboration & Leadership**
Works effectively in teams and takes the initiative to achieve project goals.
- **Project Management**
Plans, organizes, and completes tasks efficiently to meet deadlines.
- **Learning Agility**
Quickly learns new tools, technologies, and techniques to improve performance.

PROJECTS

DeepShield: Explainable Multimodal Framework for Trust Assessment and Evidence-Based Fake News Detection

(Ongoing - Expected Completion: 2027)

- Developing a multimodal AI system using text, image, and metadata for fake news detection.
- Integrating RAG and Explainable AI techniques to provide evidence-based predictions and a Trust Score.
- Building REST APIs using FastAPI and a responsive frontend using React.js.
- **Technologies:** Python, PyTorch, Transformers, BERT, CLIP, OpenCV, FAISS, FastAPI, React.js.

Medical Store Management System

June 2025 – Aug 2025

Responsibilities :

- Designed and developed responsive user interfaces using React.js.
- Created reusable React components for the web application.
- Designed the UI/UX to improve usability and user experience.
- Integrated frontend pages with the backend APIs.
- Collaborated with team members using Agile development practices.
- **Technologies:** React.js/CSS/HTML/JavaScript

End-to-End Customer Churn Prediction with MLOps

Feb 2026 – Mar 2026

Built a production-ready MLOps pipeline that predicts customer churn for Telco customers.

- Cleaned and prepared 7,043 customer records by converting data types, handling missing values, encoding categorical variables, scaling features, and creating an 80:20 train-test split.
- Trained and compared 3 machine learning models (Logistic Regression, Random Forest, and XGBoost), selecting Logistic Regression as the production model with 81.97% accuracy and 0.8621 ROC-AUC.
- **Technologies:** Python, Scikit-learn, MLflow, DVC, Apache Airflow, Docker, FastAPI, DAGsHub

SMS SPAM Detection system

Juny 2026 – July 2026

Responsibilities:

- Developed and trained a Random Forest classifier for SMS spam detection.
- Built and optimized an LSTM deep learning model for sequential text classification.
- Performed feature engineering and hyperparameter tuning to improve model performance.
- Designed and developed a responsive web-based user interface for real-time SMS spam prediction.
- Compared machine learning and deep learning model performance using Accuracy, Precision, Recall, F1Score, and ROC-AUC.
- **Technologies:** Python, Random Forest, LSTM, Flask/FastAPI, HTML, CSS, JavaScript, Scikit-learn, TensorFlow

AI-Based Fruit & Vegetable Detection Web Application

- Developed an AI-powered web application for detecting fruits and vegetables from uploaded images and live camera input.
- Integrated a Roboflow computer vision model to perform object detection and return detected classes, confidence scores, and bounding-box coordinates.
- Built a FastAPI backend with REST API endpoints for image and camera-frame detection.
- Designed a responsive and user-friendly web interface with image upload, live camera detection, loading states, and detection results.
- Implemented HTTPS support and Vercel deployment configuration for web-based access.
- **Technologies:** Python, FastAPI, Roboflow, Computer Vision, HTML, CSS, JavaScript

Hospital Appointment No-Show Prediction Using Machine Learning (Group)

August 2026

Responsibilities:

- Developed a machine learning solution to predict patient appointment no-shows using healthcare data.
- Performed data preprocessing, feature engineering, and exploratory data analysis (EDA).
- Built and evaluated a Logistic Regression classification model.
- Conducted Feature Importance Analysis to improve model interpretability.
- Developed a Streamlit web application for real-time appointment no-show prediction.
- **Technologies:** Python, Pandas, NumPy, Scikit-learn, Matplotlib, Jupyter Notebook, Streamlit, Joblib
- **Live Demo :** <https://smartcare-ai-project-11.streamlit.app/>

Mobile Application Development Project – LankaSmartMart

Jan 2026 – Feb 2026

- Designed the complete UI/UX prototype using Figma, including user-friendly screens and application workflow.
- Developed the Android mobile application using Android Studio by implementing the designed interfaces and core functionalities.
- Integrated the application with the **database** to enable real-time data storage and retrieval.
- Collaborated with team members using Git, GitHub, managing source code through version control and contributing code via commits and pushes.
- Worked closely with the development team throughout the Software Development Life Cycle (SDLC), including planning, UI/UX design, development, testing, and deployment.
- Participated in team discussions, problem-solving, and feature implementation to successfully complete the project.
- **Technologies:** Figma, Android Studio (Java/Kotlin), Firebase/MySQL, Git, GitHub

- **Demo Video:** <https://www.figma.com/proto/yQ4DUNsyD5o3vXHQSZXZ4E/Lanka-Smart-Market?node-id=2-3&t=e7TEuSFqEJ1UMqQ9-1&scaling=scale-down&content-scaling=fixed&page-id=0%3A1&starting-point-node-id=1%3A2>

REFERENCES

Available Upon Request